

WASP-36b

R-1220

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_190216 · created 2026-07-31T08:47:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458530.8055 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.048
Transit depth (Rp/Rs)^2	2.45 +/- 1.51 [%]
Orbital Inclination (inc)	81.79 +/- 2.79
Airmass coefficient 1 (a1)	0.77 +/- 0.037
Airmass coefficient 2 (a2)	0.191 +/- 0.083
Scatter in the residuals of the lightcurve fit is	4.95 %
Best Comparison Star	#1 - [227, 196]
Optimal Aperture	3.3
Optimal Annulus	18.47
Transit Duration (day)	0.06 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.048 vs 0.1368 (deviation 0.42σ)
Duration fit vs published	0.06 d vs 0.0758 d (deviation 0.55σ)
Mid-time O–C	2.1 min
Depth SNR	3.3
Residual scatter	4.95 %

Light curve

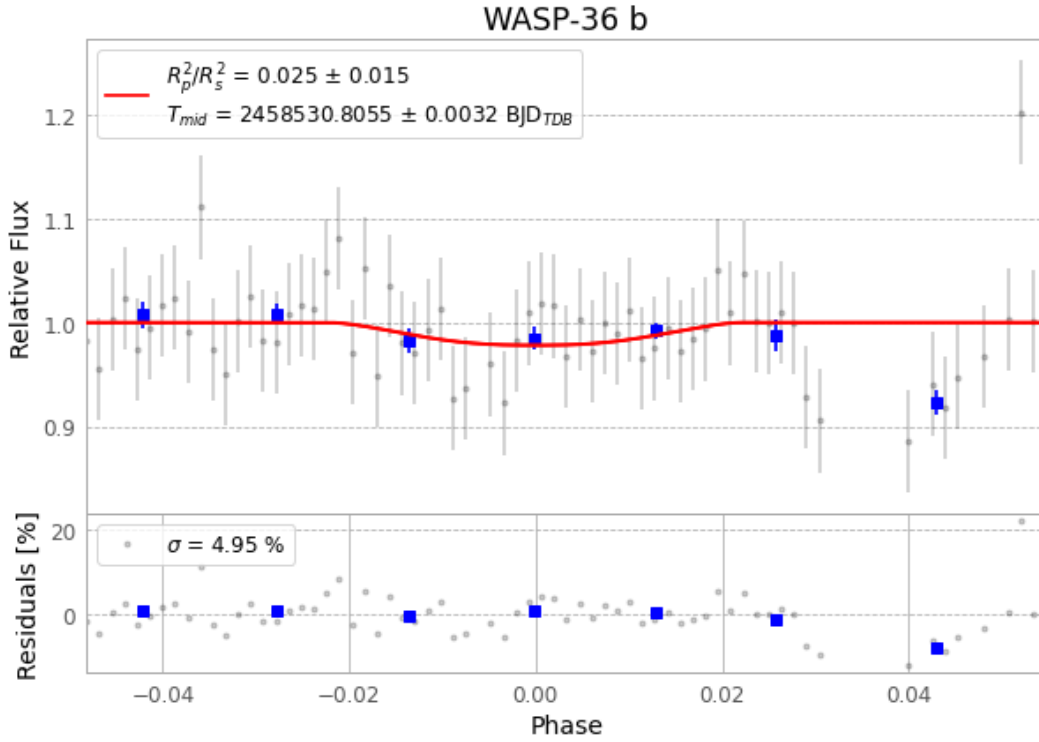


Figure 1 — FinalLightCurve_WASP-36 b_2019-02-15.png (SHA-256 3c035953c190477a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 177], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6e32e9fcb9cf8773...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36190216052412.FITS → WASP-36190216091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-190216.log (af29f13ec77d...), FinalLightCurve_WASP-36 b_2019-02-15.png (3c035953c190...), FinalLightCurve_WASP-36 b_2019-02-15.csv (39e4ff161af9...)

Compute resources

Wall time 325.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:47Z.